Simulation-Based Handover

Decision Evaluation for High-Mobility Users

Hackgen

Team ID: A3-PS016-TC274

Vaidik Upadhyay | Shourya Maheshwari

Mentor: Abhay Gupta

Team Roles & Responsibilities

Clear ownership for research, coordination, and technical development

01

Vaidik Upadhyay

TEAM LEADER & PROJECT COORDINATOR

- Overall project planning and coordination
- Problem analysis and solution planning
- Mentor communication and task management

02

Shourya Maheshwari

TECHNICAL RESEARCH & DEVELOPMENT

- GitHub and existing-solution research
- Technology stack and ML approach research
- Simulation design and technical development

03

Abhay Gupta

FACULTY MENTOR

- Technical guidance and feedback
- Review of solution design and methodology
- Guidance on evaluation and project direction

WORKING PRINCIPLE

Responsibilities describe planned ownership; simulation, testing, and validation will be reported as completed only after they are carried out.

Problem Description

PS-016 | Simulation-Based Handover Decision Evaluation for High-Mobility Users

- Mobile users move between cellular coverage areas, so the serving cell must change when needed.
- Poorly timed handovers can cause handover failures, ping-pong switching, service interruption, and reduced throughput.
- The project compares a supervised machine-learning policy with a conventional signal-based handover baseline.

  The planned simulation uses three cells, two mobility speeds, and multiple random seeds.

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Background

A handover transfers an active mobile connection from one serving cell to another as the user moves through the network.

Signal-based baseline

Uses radio measurements and configured handover rules to decide when a connection should switch cells.

Provides the conventional reference policy for the experiment.

Supervised ML policy

Learns from labelled simulation examples.

Candidate input features may include signal measurements, user speed, and recent signal changes.

Scope: simulation-based comparison only. This project does not claim a new 6G radio stack or field deployment.

Literature & Standards Review

- The problem statement requires a reproducible comparison of a learned handover policy and a signal-based baseline.
- RFC 2330, “Framework for IP Performance Metrics,” provides context for defining and reporting network performance measurements.
- The chosen simulator, radio model, software versions, configuration, and measurement method must be documented.
- Official ns-3, 5G-LENA, and scikit-learn documentation will guide implementation and reproducibility.
- The exact compatible simulator/radio-module version will be confirmed during setup; it is not presented as already implemented.

Key principle: reproducible setup + comparable scenarios + transparent reporting.

Proposed Solution

01 Configure simulator

Set up a compatible LTE/5G-capable network simulator with three cells and a moving user.

02 Collect measurements

Run two mobility speeds; collect radio measurements, handover events, interruption duration, and throughput.

03 Build baseline

Implement and validate a conventional signal-based handover baseline.

04 Train classifier

Create labelled examples and train a supervised Random Forest classifier.

05 Evaluate fairly

Evaluate baseline and ML policy on comparable scenarios across multiple random seeds.

06 Report findings

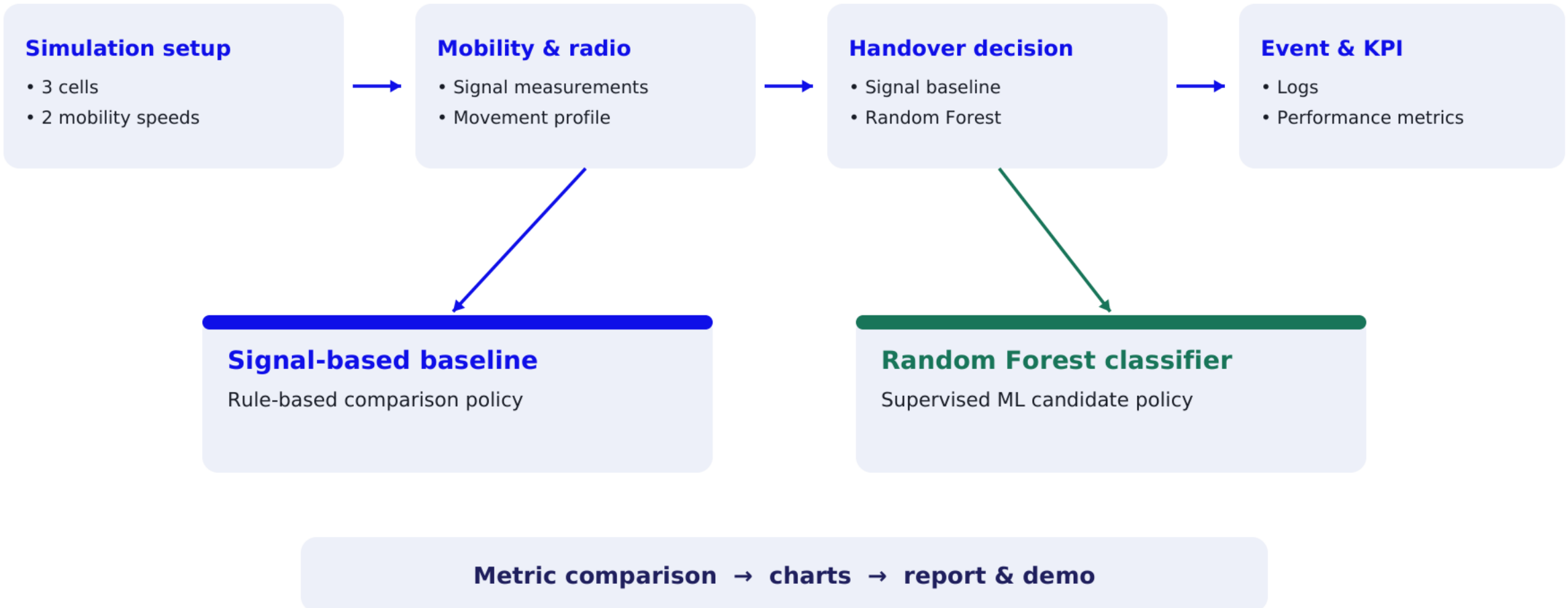
Compare metrics, visualise findings, and document settings, limitations, and demo workflow.

Solution Optimization

Performance measure	Signal-based baseline	Supervised ML policy
Handover failures	Count / rate	Count / rate
Ping-pong events	Event count	Event count
Interruption duration	Mean and distribution	Mean and distribution
Throughput	Mean and distribution	Mean and distribution

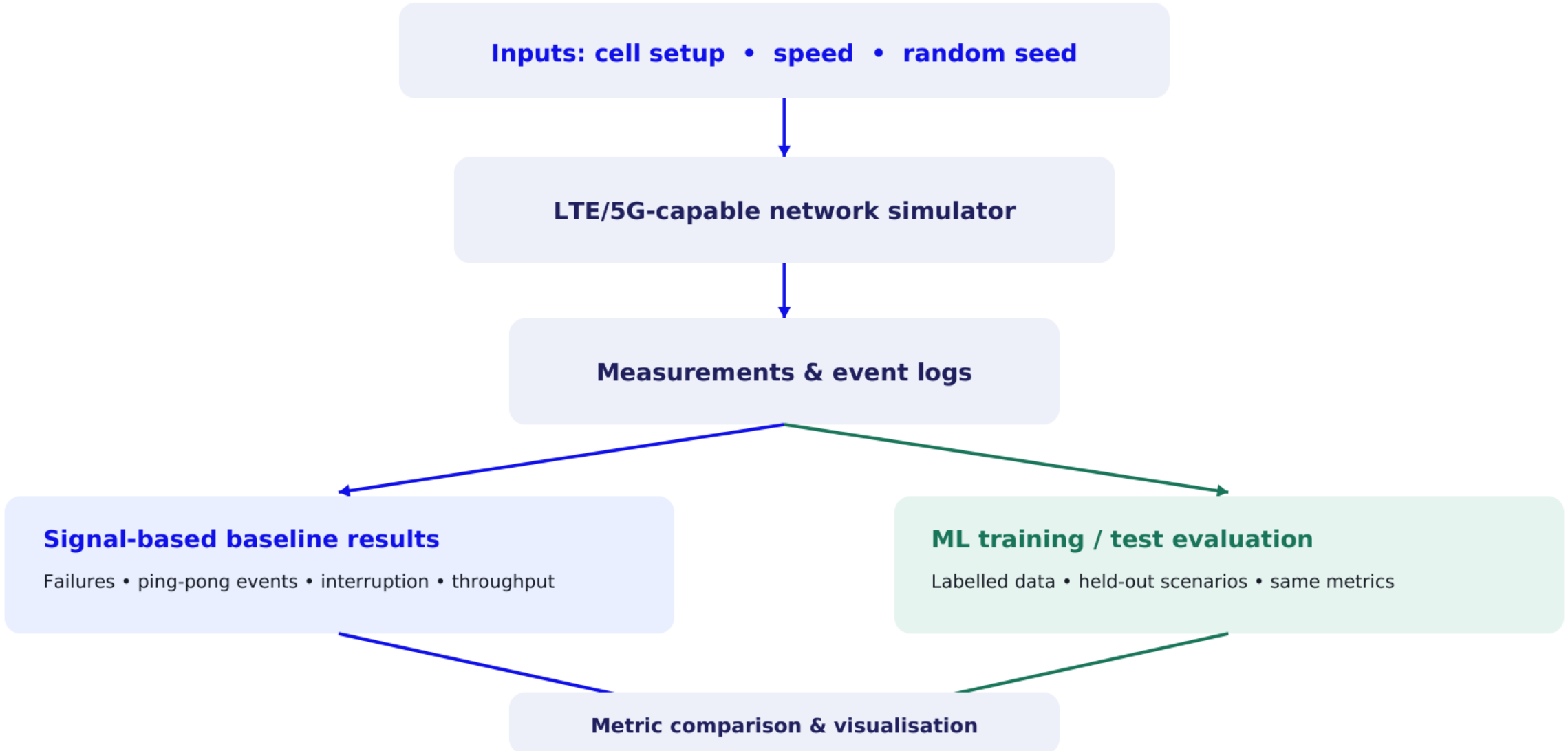
Evaluation goal: reduce failures and unnecessary switching while maintaining or improving throughput. No improvement is assumed in advance.

Solution Architecture & Design



Candidate ML model: Random Forest classifier

Data Flow Diagram



Deployment Timeline

Implementation milestones and evaluation workflow

01 **Confirm simulator and compatible radio model**

02 **Configure three cells and two mobility speeds**

03 **Implement and verify the signal-based baseline**

04 **Generate labelled data and train the ML model**

05 **Run repeated experiments with multiple random seeds**

06 **Compare four metrics; prepare charts, report, and demo**

References

1 **[1] IETF, RFC 2330, “Framework for IP Performance Metrics.”**
https://www.rfc-editor.org/rfc/rfc2330.html

2 **[2] ns-3 Network Simulator, official documentation.**
https://www.nsnam.org/

3 **[3] 5G-LENA, official project documentation.**
https://5g-lena.cttc.es/

4 **[4] scikit-learn, Random Forest and supervised learning documentation.**
https://scikit-learn.org/

Thank You

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Simulation-based handover evaluation